

The *Flight Dragon* Network

Learnable destinations, distance-priced tickets and discovery EXP for the flight mobs 47015–47041

proposed flight stop ●
named dragon · quest ◆
all 351 pairs of stops —



27

DESTINATIONS

351

ROUTES BY AIR

3

ONE-WAY FLIGHTS

Rebuilt on 24 September 2026 from the preserved text of GitHub issue #616 and its two comments, after the account that opened it was suspended.

Builder: **Fotenak** · Issue opened 22 September 2026

AT A GLANCE

27	11	12 of 16
Flight Destinations in the new design, one per flight mob, from 47015 to 47041	new flight mobs, 47031–47041, that exist in <code>vehicles.mob</code> but never load	existing flight mobs that <code>transports[]</code> still spawns at their old pads
351	12–379	3
pairs of stops, every one reachable by air under <code>VALID_FLYING_EDGE</code>	rooms flown between stops; median 201, mean 199.9	Extra Destinations: one-way flights at a fixed price, as the old system works

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THE RECORD

#616

TITLE	Flight Dragon network: learnable destinations, distance-priced tickets and discovery EXP (mobs 47015–47041)
OPENED	22 September 2026 by <code>moshehbenavraham</code>
LABELS	enhancement area:database status:deferred priority:P2 area:world area:economy
BUILDER	Fotenak, who designed the feature and built the world files
COMMENTS	2: a cross-reference to #615, and Fotenak's answers to all eleven open questions (23 September)
MOVED TO	Discussion #629 (Ideas), 23 September 2026
STATE	Issue closed as <i>not planned</i> during the move to Discussions. No code written yet.

WHY THIS DOCUMENT EXISTS

On 23 September 2026 a script moving issues to Discussions closed #616. GitHub then suspended the account that opened it, which may hide the issue text and the discussion. This record rebuilds the issue from the content backup the script took from the GitHub API just before the move (`docs/ongoing-projects/2026-09-23-issues-moved-to-discussions.md`). All of it is here: the design, the data, the investigation, and both comments.

Additions are clearly marked. They are a world map drawn from `surface.wld`, recomputed distances for Fotenak's Fort Boyard change, notes on what each answer changes, and an illustration of a list screen.



The Brief

The world files are finished but the code was never written. Eleven flight mobs never load, and most of the rest sit at the wrong pads.

NOTE AT THE TOP OF THE ISSUE

Priority: Fotenak (builder) described this as a new feature that is **not critical**. This issue records the design and the investigation so the work can be picked up later; it can be deferred.

Fotenak redesigned the flight-dragon travel system and already landed the world-file side in the vehicles area (`areas/*/vehicles.*`). The code side was never written. `src/world/transport.c` still runs the old fixed-route system from a hard-coded table, so:

- The **11 new flight mobs 47031–47041** exist in `vehicles.mob` but **never load**: nothing spawns them.
- **12 of the 16 existing flight mobs** now belong at different pads, and the builder renamed and redescribed them for their new homes. The code still spawns them at their old pads. For example, the wyvern wearing Storm Port's crest sits at *Quietus Quay*.

In the new design every flight mob is a named **Flight Destination**:

THE NEW DESIGN

- A character **learns** a destination by typing `list` at that destination's flight mob.



LEARN

A character **learns** a destination by typing `list` at that destination's flight mob.



FLY ANYWHERE

From any flight mob, the character can fly to **any destination** they know.



PAY BY DISTANCE

The **ticket price depends on the distance** between the two stops.



EXTRA DESTINATIONS

A few **Extra Destinations** keep working the way the current system does: one-way trips at a fixed price, from one specific mob.



DISCOVERY EXP

Each newly learned destination gives an **EXP reward** that **grows with the number of destinations found**.

The five points of the design, worded as in the issue's summary. The detailed behavior is in Chapter VII.

In the Builder's Words

The request as Fotenak put it in Discord, and the spreadsheet that goes with it.

“so basically when someone discovers a new flight dragon and types “list” at it for the first time, it adds that flight dragon’s “destination name” to their flight list, and now they can fly there from any other flight dragon

FOTENAK, BUILDER

THE BUILDER REPORT

AS QUOTED IN THE ISSUE

there's a whole project in the vehicles world file that is still pending coding (you might notice that like 50% of the mobs don't load)
mobs 47031-47041 (okay not 50%) i added for new destinations

well its a new feature thing so its not that critical

so the idea is that there are flight destinations that players can learn, and their cost is dynamic based on how far each one is to each other

Updated Flight Dragon system:

List of Destinations, with corresponding Destination Name, room #, corresponding mob.

Each player has a built-in list of Flight Destinations that they can travel between. New characters start with no Destinations on their list, but when they type “list” at a new flight vehicle mob, it adds that Destination to their Destination list. Ticket price based on distance between destinations.

“Extra Destinations” are 1-way trips that are hard-set price, and work like the original flight path system.

so basically when someone discovers a new flight dragon and types “list” at it for the first time, it adds that flight dragon’s “destination name” to their flight list, and now they can fly there from any other flight dragon

oh and we wanted to make it so every time you add a destination to your destinations list, it gives you an exp reward, and it goes up with the more you find (not sure how much exp, but maybe like +10k for each)

When asked, Fotenak confirmed that the original flight path system is still in the game. It is the current `transport.c` system, which dates back to at least 2009.

THE CONVERSATION, VERBATIM

Fotenak
(time cut off)

there's a whole project in the vehicles world file that is still pending coding

Fotenak
11:54 PM

(you might notice that like 50% of the mobs don't load)

Fotenak
11:55 PM mobs 47031-47041 (okay not 50%) i added for new destinations

Max aka Mosheh
11:56 PM ok want me to look into that too?

Fotenak
11:56 PM well its a new feature thing so its not that critical

Max aka Mosheh
11:56 PM whatever you need, i could create the issue now and put a note its not critical and can be put off for now too

Max aka Mosheh
11:56 PM that way there is a record of it

Fotenak
11:58 PM so the idea is that there are flight destinations that players can learn, and their cost is dynamic based on how far each one is to each other

Fotenak
11:58 PM i wrote a good couple paragraphs about it to someone let me see if i can find it

Max aka Mosheh
11:59 PM great, the more details you give, the better chance it will turn out the way you want ;p

Fotenak
11:59 PM Updated Flight Dragon system:

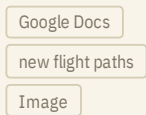
List of Destinations, with corresponding Destination Name, room #, corresponding mob.

Each player has a built-in list of Flight Destinations that they can travel between. New characters start with no Destinations on their list, but when they type "list" at a new flight vehicle mob, it adds that Destination to their Destination list. Ticket price based on distance between destinations.

"Extra Destinations" are 1-way trips that are hard-set price, and work like the original flight path system.

Max aka Mosheh
12:00 AM there is a flight path system? it was taken out? you happen to remember how long ago it existed?

Fotenak
12:00 AM there is a spreadsheet that accompanies this:
https://docs.google.com/spreadsheets/d/19gZNDKk_n4MWBtkpmNHXTmnLNErQh0SaAFEap2dbsA/edit?usp=drive_link



Google Docs

new flight paths

Image

Fotenak
12:01 AM its still in

Max aka Mosheh
12:02 AM ok

Fotenak
12:03 AM so basically when someone discovers a new flight dragon and types "list" at it for the first time, it adds that flight dragon's "destination name" to their flight list, and now they can fly there from any other flight dragon

Fotenak
12:13 AM oh and we wanted to make it so every time you add a destination to your destinations list, it gives you an exp reward, and it goes up with the more you find

Fotenak
12:13 AM (not sure how much exp, but maybe like +10k for each)

The first line of the transcript starts partway through its timestamp (**PM**Fotenak:); it is shown as it was pasted. Max aka Mosheh is the maintainer who opened the issue.

THE SPREADSHEET

The data comes from the spreadsheet [new flight paths](#). The "Image" in the Discord post appears to be the link's preview image. The sheet has **two tabs**:

- **NEW FLIGHT VEHICLES** (gid 1897554044) is the new design. Its CSV export is identical, apart from line endings, to the CSV we were given (`tmp/new-flight-paths.csv`). Every value is reproduced below.
- **old routes** (gid 0) documents the current system. Its 20 rows match today's `transport_routes[]` row for row: the same pads, destinations, prices and level gates. It ends with one extra row that has only a destination name, "**Behemoth Herders**" (see the open questions). Reproduced verbatim at the end of this section.

The Atlas of Flight

● Flight Destination ◆ Named dragon (Quest? = Y) - - ◆ Extra Destination (one-way)



The 27 stops of the proposed network on the surface map (zone #5000, rooms 500000–659999). The map is a 400 × 400 grid where $vnum = 500000 + 400 \cdot y + x$. It wraps on both axes, so a flight can leave one edge and come in on the opposite one. Stops in bold are the seven named dragons. Grid lines are every 50 rooms. Terrain is drawn from each room's sector type in `areas/wld/surface.wld`, with relief shading from hills and mountains.

Shown as amended by Fotenak's answers of 23 September. Fort Boyard's pad is 563522, and 47038's stop is labeled *Dalvik's Dockyard* until *Manshaka* is built. Violet arrows are the three Extra Destinations: Chief Blood-Eye to the Bloodyfist cavern, the Isle of Undeath to A Desolate Island, and Storm Port to Storm Port Stronghold.

The Register

Each Flight Destination with its mob, pad, name and in-game colors, followed by the one-way Extra Destinations.

The spreadsheet tab *NEW FLIGHT VEHICLES* lists every destination: the mob that serves it, the pad room it loads in, and a destination name in plain text and with ANSI colors. Below, names are drawn in their in-game colors. The *flight mob* column shows the short description as it stands in `areas/mob/vehicles.mob` today.

DESTINATIONS (27)

MOB	DESTINATION NAME (IN GAME)	PAD ROOM	X, Y	FLIGHT MOB IN VEHICLES.MOB	OLD MOB NAME
47027 ♦	Swamp of Chief Blood-Eye	619771	(171, 299)	Krrlytstalarryn, the ancient red dragon	Krrlytstalarryn, the ancient red dragon
47015 ♦	The Isle of Undeath	576577	(177, 191)	Corrosyville, the ancient black dragon	huge skeletal dragon
47016	Quietus Quay	635756	(156, 339)	a huge wyvern	mighty gryphon
47017	Storm Port	588326	(326, 220)	a mailed wyvern	a giant green dragon
47018	Bloodstone Keep	606262	(262, 265)	an armored wyvern	huge hippogryph
47019	Myrabolus	514072	(72, 35)	a great white hippogriff	flying carpet
47020	The City of Torrhan	582084	(84, 205)	a great white griffin	emerald dragon
47021	Ugta	543701	(101, 109)	a huge griffin	powerful gryphon
47022	Venan'Trut	545407	(207, 113)	a flying carpet	hippogryph
47023 ♦	Fort Boyard	564319 563522 AMENDED	(319, 160) (322, 158)	Azerathrax, the ancient blue dragon	onyx dragon
47024	Sarmiz'Duul	622276	(276, 305)	a large wyvern	a great blue drake
47025	Fort Khoralator	519043	(243, 47)	a harnessed roc	a saddled fire drake
47026	Kimordril	543662	(62, 109)	a harnessed griffin	a GREAT wyvern
47028	Helgor Outpost	522247	(247, 55)	a harnessed frost drake	a mighty griffon
47029	Clan Shatter Stone	631413	(213, 328)	a harnessed wyvern	a lithe black drake
47030 ♦	Bone Fort Inn	514279	(279, 35)	Cryoss, the ancient white dragon	a saddled white drake
47031	Winterhaven	533479	(279, 83)	a mighty roc	new mob
47032	Shady Grove	584293	(293, 210)	a mighty wyvern	new mob
47033	Tharnadia	570024	(24, 175)	a mighty griffin	new mob
47034	Fenaline	529229	(29, 73)	a large hippogriff	new mob
47035	Canderthal Harbor	546144	(144, 115)	a large griffin	new mob
47036 ♦	Nizari	575328	(128, 188)	Aurumax, the ancient copper dragon	new mob
47037 ♦	The Jade Empire	641295	(95, 353)	Verdax, the ancient jade dragon	new mob

MOB	DESTINATION NAME (IN GAME)	PAD ROOM	X, Y	FLIGHT MOB IN VEHICLES.MOB	OLD MOB NAME
47038	Manshaka SHOWN AS DALVIK'S DOCKYARD FOR NOW	634060	(60, 335)	a mighty great eagle	new mob
47039	Varathorn Keep	577835	(235, 194)	a huge skeletal dragon	new mob
47040	Thur'Gurax	616547	(147, 291)	a large wyvern	new mob
47041 ♦	Moonshae Island	557220	(20, 143)	Glimmyris, the ancient faerie dragon	new mob

♦ marks *Quest?* = *Y* on the sheet: the seven named dragons Krrlytstalarryn, Corrosyvile, Azerathrax, Cryoss, Aurumax, Verdax and Glimmyris. The sheet ends with an empty placeholder row for **47042**. The sheet originally gave "a great white giant eagle" for 47038 and "Glimmyris, the great faerie dragon" for 47041. Fotenak has since changed the sheet to match the mob file (answer 9), so the mob-file names above are the canonical ones.

WHERE EACH PAD SITS

MOB	DESTINATION	PAD	ROOM TITLE	NEAREST MATCHING ZONE ENTRANCE
47027	Swamp of Chief Blood-Eye	619771	A Rugged and Barren Hillside	Map region (The Foul Bog of Chief Bloodeye). Valley of the Snow Ogres (snogres) entrance is 36 rooms away
47015	The Isle of Undeath	576577	An Ominous Landing	Map region (Lifeless Plains of Dead Rising, Jungle of the Living Dead). No zone entrance within 6 rooms
47016	Quietus Quay	635756	A Marshy Clearing with Patches of Tanglebush	quietus #17, 1 room
47017	Storm Port	588326	Rugged Grasslands Along the Coast	stormport #224, 3 rooms (south gate 22401)
47018	Bloodstone Keep	606262	A Rocky Climb Along the DarkPeak Mountains	bs #740, 3 rooms
47019	Myrabolus	514072	Northern Valley of Myrabolus	mira #825, 3 rooms
47020	The City of Torrhan	582084	Endless Plains of Grassy Pastures	torrhan #666, 6 rooms
47021	Ugta	543701	Endless Plains of Grassy Pastures	ugta #391, 4 rooms
47022	Venan'Trut	545407	The Amber Sands of an Island Desert	desert #490, 13 rooms
47023	Fort Boyard	564319 563522	A Beach of Fine Sand Foothills of the Mountains of Ciuto	fortb #381, 6 rooms
47024	Sarmiz'Duul	622276	Vast Grasslands Within the DarkPeak Valley	sarmiz #94, 2 rooms
47025	Fort Khoralator	519043	A Spacious Mountain Clearing	alatorin #831, 1 room
47026	Kimordril	543662	A Rocky Mountain Clearing	kimordril #955, 1 room
47028	Helgor Outpost	522247	A Mountain Clearing	alatorin #831, 1 room
47029	Clan Shatter Stone	631413	A Small Covered Mountain Outpost	kvarkpass_connector #546, 1 room
47030	Bone Fort Inn	514279	The Frozen Grassland of the Northern Tundra	tundra #137, 1 room (Bone Fort Inn rooms 13712+)
47031	Winterhaven	533479	Lush Plains of Shrubs and Small Bushes	wh #550, 1 room
47032	Shady Grove	584293	Rugged Hillside of the Northern Wilds	shady #975, 1 room
47033	Tharnadia	570024	A Recently Rebuilt Stone Highway	tharnadia #1325, 1 room
47034	Fenaline	529229	The Lowlands of a Coastal Rim	shipy #431 (Fenaline Shipyard 43181+), 2 rooms
47035	Canderthal Harbor	546144	A Sloped and Rocky Hillside	shipy #431 (Canderthal Harbor 43101+), 2 rooms
47036	Nizari	575328	Sand Dunes of the Calimshan Desert	nizari #400, 1 room
47037	The Jade Empire	641295	The Jade Hills	jademini #772, 3 rooms; jade #766, 6 rooms
47038	Manshaka	634060	A Tropical Sandy Beach	None built. dalvik is not in areas/AREA; its Dalvik's Bay link at 635660 is 4 rooms away
47039	Varathorn Keep	577835	Rocky Foothills of the Black Mountains	kastle #994, 4 rooms

MOB	DESTINATION	PAD	ROOM TITLE	NEAREST MATCHING ZONE ENTRANCE
47040	Thur'Gurax	616547	The Dreary and Rocky Hillside of Chief Bloodeye	shipy #431 (Thur'Gurax Port 43141+), 1 room
47041	Moonshae Island	557220	The Gentle Rolling Hills of Moonshae Island	moonshae #262, 2 rooms

Zone entrances are from the issue's verification against `areas/world.wld`. The room title for the amended Fort Boyard pad, 563522, was read from `surface.wld` for this record.

EXTRA DESTINATIONS

Extra Destinations are one-way trips at a fixed price. Each departs from one specific mob and works the way the original flight path system does.

FROM	TO ROOM	NAME (IN GAME)	MIN LEVEL	COST	ROOMS FLOWN	ROOM TITLE	TODAY
47027 Swamp of Chief Blood-Eye (619771)	620606	A Huge Cavern on the Bloodyfist Mountains	51	100	37	An Entrance to a Huge Cavern	50p, level 51+
47015 The Isle of Undeath (576577)	579365	A Desolate Island	51	100	19	The Lifeless Plains of a Desolate Island	200p, level 51+
47017 Storm Port (588326)	637082	Storm Port Stronghold	1	10	166	A Patch of Grass	none (new route)

The sheet gives no currency for "Extra Destination Cost". The current code prices every route in platinum, so these are assumed to be platinum. Fotenak's answer 4 removes restrictions "to start", which may also cover these level gates; see Chapter VIII.

BUILDER-SIDE DATA NITS

NIT, AS RECORDED	STATUS AFTER FOTENAK'S ANSWERS
47038 name. The sheet says "a great white giant eagle"; the mob file says "a mighty great eagle".	Sheet now matches the mob file (answer 9). RESOLVED
47041 name. The sheet says "Glimmyris, the great faerie dragon"; the mob file says "Glimmyris, the ancient faerie dragon".	Sheet now matches the mob file (answer 9). RESOLVED
47015 description. It was renamed "Corrosyville, the ancient black dragon", but its detailed description still describes the old skeleton ("This massive skeleton is that of a dragon..."). The skeletal-dragon role now belongs to 47039 at Varathorn Keep.	Fotenak will expand the description with the quest work (answer 9). PENDING
47018 typo. The long description says "adnored".	Not addressed in the answers. OPEN
47034 ANSI name. &+cFenaline has no closing &N. Output code should always terminate names anyway.	Not addressed in the answers. OPEN
Extra Destination price changes. The Snow Ogres cavern (620606) goes from 50p to 100p. The Desolate Island (579365) goes from 200p to 100p. Storm Port → Stronghold is new: 10p, level 1+.	The prices stand. Answer 4 removes restrictions to start. CHANGED



The Service Today

How `src/world/transport.c` runs the 2009 fixed-route service, and how far the world files have moved away from it.

All links point at master `a71fdfee7`.

1

Two hard-coded tables drive the service.

`transport_routes[]` holds 20 one-way routes. Each has an origin room, a destination room, a display name, a price in platinum, and a min/max level. `transports[]` maps 16 mob vnums to their origin rooms.

2

Boot.

`initialize_transport()` runs once during startup, after file copyover or Redis recovery (or the Redis fallback) has restored the world. It:

- precomputes each route's path with `dijkstra(..., valid_flying_edge, ...)`, printing `no route found` to stderr when that fails;
- attaches the `flying_transport` proc to each listed mob;
- keeps exactly one dragon per listed mob, spawning a missing one at its origin and retiring duplicates;
- makes sure each origin has one `sign, obj 420`.

3

list.

`flying_transport_cmd_list` shows the routes whose origin is the mob's current room and whose level gates the player passes. `IS_TRUSTED` bypasses the gates.

4

buy <n>.

`flying_transport_cmd_buy` charges `cost_in_plat * 1000` coins. It hands over `ticket obj 47008` with `value[6] = mob vnum` and `value[7] = index into transport_routes[]`.

5

give ticket <mob>.

`flying_transport_cmd_give` consumes the ticket, mounts the player (`LNK_RIDING`) and starts `event_flying_transport_move`. The dragon flies the precomputed path one room per event. At the destination the rider dismounts, and the dragon flies home along the reversed path.

6

While flying, the rider can only use `look`, `score`, `inventory`, `attributes`, `news`, `petition` and `toggle` (`L609-L618`). Everything else gets "you're holding on for dear life!".

7

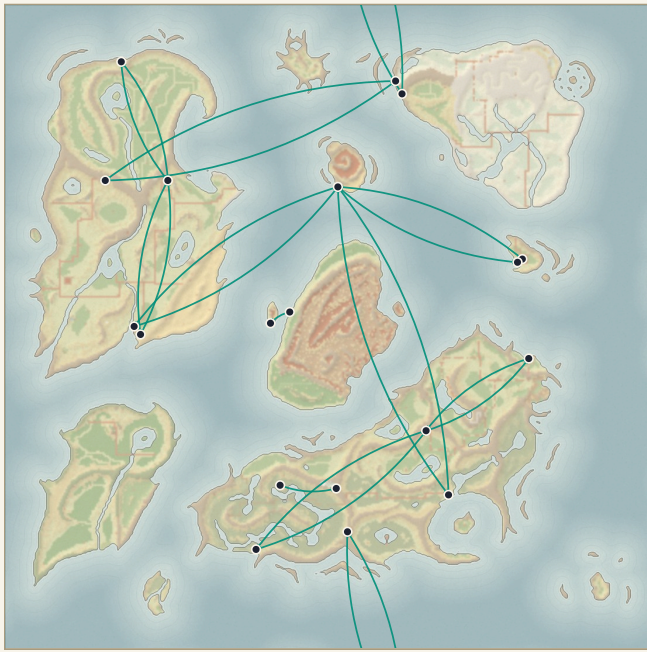
Flight edges.

`VALID_FLYING_EDGE` needs a two-way exit inside the **same zone** that is not `EX_BLOCKED`. Every stop in this project is on the surface map: zone #5000, rooms 500000–659999, a 400×400 grid where `vnum = 500000 + 400·y + x`. The grid **wraps on both axes**: room 500000's north exit leads to 659600 and its west exit to 500399.

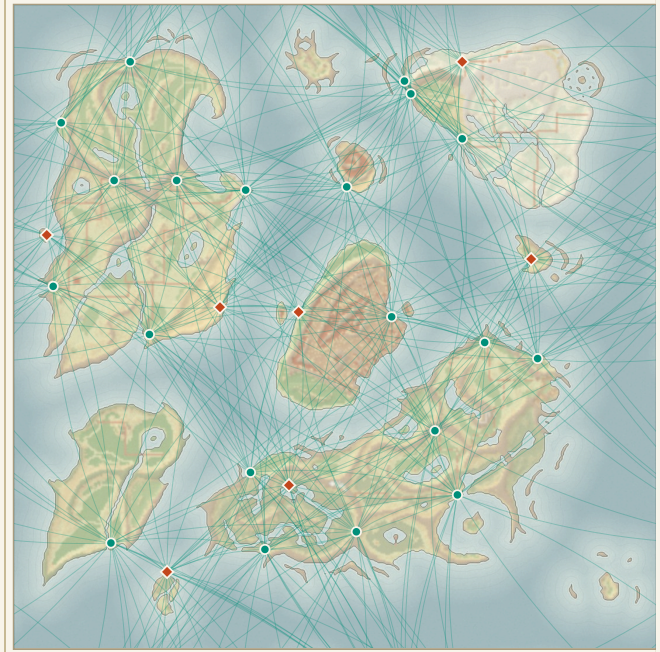
Recovery.

`transport_snapshot` stores the origin and destination **room vnums**, the state, the step and the rider's name. It is used for copyover v13 and Redis schema 12 (TRN1 tail). `transport_restore()` finds the route again from (origin, destination). If a snapshot's origin no longer matches the configured origin, `transport_land_at_home()` brings the dragon home. See [docs/operations/WORLD_SINGLETON_RECOVERY.md](#) (#235, #315).

THE TWENTY FIXED ROUTES



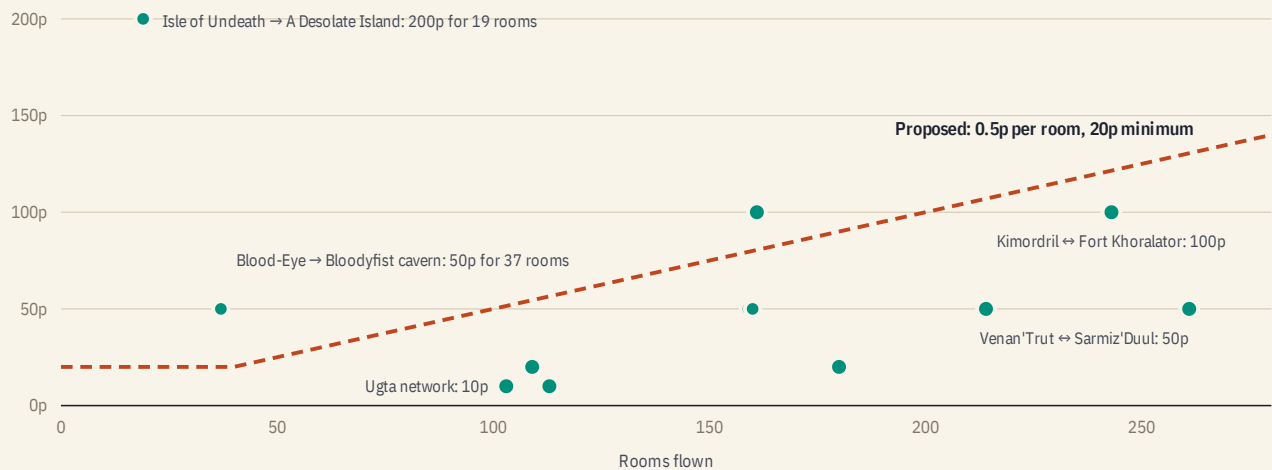
Today: 20 one-way routes between 16 pads, from `transport_routes[]`.



Proposed: 27 stops, any known stop reachable from any other. All 351 pairs are drawn.

#	MOB SERVING THE ORIGIN TODAY	FROM	TO	PRICE	LEVELS	ROOMS FLOWN	PLATINUM PER ROOM	
0	47027	619771 (Krrlytstalarryn pad)	620606 A Huge Cavern on the Bloodfist Mountains	50p	51+	37	1.35	■
1	47015	576577 (skeletal dragon pad)	579365 A Desolate Island pad	200p	51+	19	10.53	■
2	47016	606262 Bloodstone Keep	635756 Quietus Quay	20p	—	180	0.11	└
3	47016	606262 Bloodstone Keep	588326 Storm Port	20p	—	109	0.18	└
4	47017	635756 Quietus Quay	606262 Bloodstone Keep	20p	—	180	0.11	└
5	47018	588326 Storm Port	606262 Bloodstone Keep	20p	—	109	0.18	└
6	47021	543701 Ugta	514072 Myrabolus	10p	—	103	0.10	└
7	47021	543701 Ugta	582084 The City of Torrhan	10p	—	113	0.09	└
8	47025	582084 Torrhan	543701 Ugta	10p	—	113	0.09	└
9	47022	514072 Myrabolus	543701 Ugta	10p	—	103	0.10	└
10	47020	580080 Torrhan (2nd pad)	545407 Venan'Trut	50p	47+	214	0.23	└
11	47019	545407 Venan'Trut	564319 Fort Boyard	50p	46+	159	0.31	└
12	47019	545407 Venan'Trut	622276 Sarmiz'Duul	50p	46+	261	0.19	└

	MOB SERVING THE # ORIGIN TODAY	FROM	TO	PRICE LEVELS	ROOMS FLOWN	PLATINUM PER ROOM
13	47019	545407 Venan'Trut	580080 The City of Torrhan (2nd pad)	50p 46+	214	0.23 ↑
14	47023	622276 Sarmiz'Duul	545407 Venan'Trut	50p 47+	261	0.19 ↑
15	47024	563522 Fort Boyard (old pad)	545407 Venan'Trut	50p 47+	160	0.31 ↑
16	47030	543662 Kimordril	519043 Fort Khoralator	100p —	243	0.41 ↑
17	47028	519043 Fort Khoralator	543662 Kimordril	100p —	243	0.41 ↑
18	47029	631413 Clan Shatter Stone	522247 Helgor Outpost	100p —	161	0.62 ↑
19	47026	522247 Helgor Outpost	631413 Clan Shatter Stone	100p —	161	0.62 ↑



Today's prices have no relation to distance. They range from 0.09 to 10.5 platinum per room flown. Each dot is a current route (pairs flown both ways overlap). The dashed line shows the rule from Fotenak's answer 1: 0.5p per room with a 20p minimum.

WHY 47031-47041 DON'T LOAD

- `transports[]` has no entries for them, so `initialize_transport()` never spawns them and never attaches the proc.
- No zone file has an `M` reset for any vnum from 47015 to 47042. That part is expected: code spawns flight mobs, not zone resets.

THE PLACEMENTS HAVE DRIFTED

4ee13487a (2026-06-11) renamed and redescrbed 15 of the 16 mobs from 47015 to 47030 (all except 47027) and added 47031-47041. It re-landed "zone updates from fotenak", which first went in as 0fa96b480 and was reverted in 1538cfcc4. `transports[]` was never updated:

MOB	NAME IN THE MOB FILE NOW	SPAWNED TODAY AT (<code>TRANSPORTS[]</code>)	SHEET LOAD ROOM	RESULT
47027	Krrlystalarryn, the ancient red dragon	619771	619771 Swamp of Chief Blood-Eye	same pad
47015	Corrosyvile, the ancient black dragon	576577	576577 The Isle of Undeath	same pad (renamed)
47016	a huge wyvern	606262 Bloodstone Keep	635756 Quietus Quay	moves
47017	a mailed wyvern (Storm Port crest)	635756 Quietus Quay	588326 Storm Port	moves
47018	an armored wyvern (Bloodstone Keep banners)	588326 Storm Port	606262 Bloodstone Keep	moves

MOB	NAME IN THE MOB FILE NOW	SPAWNED TODAY AT (TRANSPORTS[])	SHEET LOAD ROOM	RESULT
47019	a great white hippogriff	545407 Venan'Trut	514072 Myrabolus	moves
47020	a great white griffin	580080 Torrhan (2nd pad)	582084 The City of Torrhan	moves
47021	a huge griffin	543701 Ugta	543701 Ugta	same pad (renamed)
47022	a flying carpet	514072 Myrabolus	545407 Venan'Trut	moves
47023	Azerathrax, the ancient blue dragon	622276 Sarmiz'Duul	564319 Fort Boyard 563522 Fort Boyard AMENDED	moves
47024	a large wyvern	563522 Fort Boyard (old pad)	622276 Sarmiz'Duul	moves
47025	a harnessed roc	582084 Torrhan	519043 Fort Khoralator	moves
47026	a harnessed griffin	522247 Helgor Outpost	543662 Kimordril	moves
47028	a harnessed frost drake	519043 Fort Khoralator	522247 Helgor Outpost	moves
47029	a harnessed wyvern	631413 Clan Shatter Stone	631413 Clan Shatter Stone	same pad (renamed)
47030	Cryoss, the ancient white dragon	543662 Kimordril	514279 Bone Fort Inn	moves (new stop)
47031– 47041	11 new mobs	never spawned	new stops (see sheet)	missing



The drift, drawn

Each arrow runs from where `transports[]` spawns a mob today to the pad the sheet gives it. Labels are the last two digits of the mob vnum (470xx). 12 of the 16 existing mobs move. Dotted teal rings are the eleven stops for 47031–47041, which never spawn.

- Spawned there today
- Pad in the new design
- ⦿ New stop, never spawned

Visible in game today:

- The "mailed wyvern" whose flanks bear "the gale-crest of Storm Port" (47017) serves *Quietus Quay*.
- The "armored wyvern" with Bloodstone Keep banners (47018) serves Storm Port.
- Cryoss (47030) serves Kimordril.

The new plan retires two pads:

- **580080**, the second Torrhan pad used by the Venan'Trut network.
- **563522**, the old Fort Boyard departure pad. Arrivals already landed at 564319, the new Fort Boyard pad.

Amended: Fotenak moved Fort Boyard back to 563522 (answer 10), so only 580080 retires.

Proof by Air

Every value in the sheet, checked against the world the server actually builds.

I checked every sheet value against the world the server actually builds, `areas/world.wld`, using read-only scripts. `areas/AREA` builds `surface`. The stale copies `newbiemaps.wld` and `surface2011-temp.wld` also define some of these vnums, but they are not built.

✓ All 27 load rooms and all 3 Extra Destination rooms exist.

All are in zone #5000 (`surface.wld`).

✓ Every pair of stops is reachable by air.

I checked all 351 pairs under the exact `VALID_FLYING_EDGE` rule, with 0 failures. `surface.zon` has no `D` resets, so no surface exit starts out blocked.

✓ Distances.

The shortest hop is 12 rooms (Fort Khoralator ↔ Helgor Outpost). The median is 201, the mean 199.9 and the longest 379 (Sarmiz'Duul ↔ Kimordril). 15 pairs are 60 rooms or less, and 40 pairs are 300 or more.

✓ Path length vs. map distance.

For every pair, the flight path length is within 3 rooms of the wrap-around Manhattan distance on the 400×400 grid, so pricing by path length or by coordinates gives the same result. A coordinate formula **must take the shorter way around each axis**, because many shortest flights cross the edge of the map.

✓ Pads sit at their towns.

Most load rooms are 1–6 rooms from the matching zone's map entrance (see the table below).

✓ Storm Port Stronghold (637082) is correct.

It is directly south of 636682, the map room that zone #226 "Storm Port Stronghold" (`spshold`) connects to. That zone is 166 flight rooms away from Storm Port itself.

! Manshaka (634060) has no town in the game yet.

- The Manshaka rooms live in `areas/wld/dalvik.wld` ("Dalvik's Dockyard").
- That file is **not in** `areas/AREA`.
- It still uses placeholder vnums #1–#476, which collide with existing rooms (for example, #1 is Limbo).
- Its only map link is Dalvik's Bay → 635660, at (60,339). That is 4 rooms from the Manshaka pad at (60,335).
- Until the zone is renumbered and added to the build, a Manshaka flight lands on "A Tropical Sandy Beach" in Khomani-Khan with no town nearby.

✓ The shipyard ports are real.

Thur'Gurax, Fenaline and Canderthal Harbor are ports in the Ship Yards zone (`shipy`, #431). Their rooms are "Thur'Gurax Port" 43141+, "Fenaline Shipyard" 43181+ and "Canderthal Harbor" 43101+. Each pad is 1–2 rooms from a Ship Yards map entrance.

i Destination names are labels, not room titles.

Chief Blood-Eye is a region on the map ("The Foul Bog of Chief Bloodeye"; 2,759 map rooms carry Bloodeye names), not a zone. The Isle of Undeath is also a map region. For example, 619771's room title is "A Rugged and Barren Hillside".

- ✓ **Every flight mob stays put.**
All 27 are `ACT_SENTINEL` and none has `ACT_SPEC`. The proc is attached in code, and the service only answers while the mob is in its origin room.
- ! **No flight mob has a `.qst` entry,**
although the sheet marks 7 named dragons `Quest? = Y` (see the open questions).
- ✓ **No stop blocks EXP.**
None of the 27 stops is `ROOM_SAFE`, `ROOM_GUILD` or `ROOM_ARENA`, so `gain_exp()` will not zero a discovery reward there. The only flags present are Kimordril's `ROOM_NO_TELEPORT` and Fort Boyard's `ROOM_DOCKABLE`.

12	201	379
rooms: the shortest hop, Fort Khoralator ↔ Helgor Outpost	rooms: the median hop (mean 199.9)	rooms: the longest hop, Sarmiz'Duul ↔ Kimordril

THE DISTANCE MATRIX

Rooms flown between every pair of stops, computed with the `VALID_FLYING_EDGE` rule. Rows and columns are the destinations, labeled with the last two digits of the mob vnum (470xx). The matrix is symmetric. **The last row is new:** it is Fort Boyard measured from 563522, the pad Fotenak moved it back to. That pad is 5 rooms from the old one, so each distance changes by 5 rooms or less.

	Chief Blood-Eye	Isle of Undeath	Quietus Quay	Storm Port	Bloodstone Keep	Myrabolus	Torrhan	Ugta	Venan'Trut	Fort Boyard	Sarmiz'Duul	Fort Khoralator	Kimordril	Helgor Outpost	Clan Shatter Stone	Bone Fort Inn	Winterhaven	Shady Grove	Tharnadia	Fenaline	Canderthal Harbor	Nizari	Jade Empire	Manshaka	Varathorn Keep	Thur'Gurax	Moonshae Island
27 Chief Blood-Eye	-	114	55	234	125	235	181	260	222	284	111	220	299	232	71	244	292	211	271	316	211	154	130	147	169	32	307
15 Isle of Undeath	114	-	169	177	159	261	107	158	108	173	213	210	197	206	173	258	210	135	169	266	109	52	244	261	61	130	205
16 Quietus Quay	55	169	-	289	180	180	206	225	225	339	154	195	264	207	68	219	267	266	296	261	188	179	75	100	224	57	332
17 Storm Port	234	177	289	-	109	331	173	286	225	67	135	255	247	243	221	231	183	43	143	250	286	229	302	249	116	250	171
18 Bloodstone Keep	125	159	180	109	-	360	238	317	207	159	54	201	353	205	112	187	199	86	249	356	268	211	255	268	98	141	277
19 Myrabolus	235	261	180	331	360	-	182	103	213	278	326	183	84	195	248	193	241	351	188	81	152	209	105	112	322	219	160
20 Torrhan	181	107	206	173	238	182	-	113	215	210	292	317	118	313	252	365	317	193	90	187	150	61	159	154	162	149	126
21 Ugta	260	158	225	286	317	103	113	-	110	233	371	204	39	200	293	252	204	293	143	108	49	106	162	215	219	228	115
22 Venan'Trut	222	108	225	225	207	213	215	110	-	159	261	102	149	98	191	150	102	183	245	218	65	154	272	325	109	238	217
23 Fort Boyard	284	173	339	67	159	278	210	233	159	-	185	189	194	177	271	165	117	73	120	197	220	219	369	316	118	300	118
24 Sarmiz'Duul	111	213	154	135	54	326	292	371	261	185	-	175	379	179	86	133	181	112	275	321	322	265	229	214	152	143	303
25 Fort Khoralator	220	210	195	255	201	183	317	204	102	189	175	-	243	12	149	48	72	213	309	212	167	256	242	295	155	252	273
26 Kimordril	299	197	264	247	353	84	118	39	149	194	379	243	-	239	332	257	209	267	104	69	88	145	189	176	258	267	76
28 Helgor Outpost	232	206	207	243	205	195	313	200	98	177	179	12	239	-	161	52	60	201	297	200	163	252	254	307	151	264	261
29 Clan Shatter Stone	71	173	68	221	112	248	252	293	191	271	86	149	332	161	-	173	221	198	342	329	256	225	143	160	156	103	378
30 Bone Fort Inn	244	258	219	231	187	193	365	252	150	165	133	48	257	52	173	-	48	189	285	188	215	304	266	281	203	276	249
31 Winterhaven	292	210	267	183	199	241	317	204	102	117	181	72	209	60	221	48	-	141	237	160	167	256	314	329	155	324	201
32 Shady Grove	211	135	266	43	86	351	193	293	183	73	112	213	267	201	198	189	141	-	163	270	244	187	341	292	74	227	191
33 Tharnadia	271	169	296	143	249	188	90	143	245	120	275	309	104	297	342	285	237	163	-	107	180	117	249	196	208	239	36
34 Fenaline	316	266	261	250	356	81	187	108	218	197	321	212	69	200	329	188	160	270	107	-	157	214	186	169	315	300	79
35 Canderthal Harbor	211	109	188	286	268	152	150	49	65	220	322	167	88	163	256	215	167	244	180	157	-	89	211	264	170	179	152
36 Nizari	154	52	179	229	211	209	61	106	154	219	265	256	145	252	225	304	256	187	117	214	89	-	198	215	113	122	153
37 Jade Empire	130	244	75	302	255	105	159	162	272	369	229	242	189	254	143	266	314	341	249	186	211	198	-	53	299	114	265
38 Manshaka	147	261	100	249	268	112	154	215	325	316	214	295	176	307	160	281	329	292	196	169	264	215	53	-	316	131	232
39 Varathorn Keep	169	61	224	116	98	322	162	219	109	118	152	155	258	151	156	203	155	74	208	315	170	113	299	316	-	185	236
40 Thur'Gurax	32	130	57	250	141	219	149	228	238	300	143	252	267	264	103	276	324	227	239	300	179	122	114	131	185	-	275
41 Moonshae Island	307	205	332	171	277	160	126	115	217	118	303	273	76	261	378	249	201	191	36	79	152	153	265	232	236	275	-
23 Fort Boyard at 563522	289	178	344	66	164	273	209	228	160	5	190	190	189	178	276	166	118	78	119	192	221	224	368	315	123	305	113
Rooms flown	12	60	110	160	210	260	310	379																			

The original matrix is reproduced exactly from the issue. For this record it was recomputed from `surface.wld` by breadth-first search, and all 702 directed cells matched (Appendix B).

What to Build

The requested behavior, the implementation notes, the tests, and the suggested phasing, as recorded in the issue.

§1 REQUESTED BEHAVIOR

1

Destination catalog.

The 27 stops from the sheet. Each has a mob vnum, a pad room, a destination name and an ANSI name.

2

Learning.

- New characters know no destinations.
- The first `list` at a flight mob adds that mob's destination, with a message.
- The list persists across logout, reboot and copyover.

3

`list` at a flight mob shows:

- every known destination except the current one, priced by distance from this stop;
- this mob's Extra Destinations: one-way, fixed price, level-gated, as today.

4

Buying and flying.

`buy <n>` buys a ticket for that entry. `give ticket <mob>` flies there, as today.

5

Discovery EXP.

Each newly learned destination awards EXP. The amount grows with how many destinations the character has found (roughly +10k per step; see below).

§2 CATALOG AND PATHS

Replace the two tables.

A destination catalog plus an Extra Destination table replaces `transports[]` and `transport_routes[]`. Static tables in `transport.c` match current practice; the ferries are also a `static table in ferryact.c`.

Paths: one search per stop, not one per route.

27 stops make 702 directed pairs. The current per-route `dijkstra()` allocates two vectors the size of `top_of_world` on every call and explores outward until it reaches the target. Instead:

- run one single-source search per stop at boot (27 searches);
- use plain BFS, because every edge costs 1;
- store the direction list for each destination; the return trip is the reverse;
- or compute paths lazily and cache them.
- Keep logging and dropping any unreachable pair, as today.

Mob route state.

`TRANSPORT_ROUTE(mob)` (`npc value[2]`) is an index into `transport_routes[]` today. It needs to identify the destination (or the origin/destination pair) instead, with the path looked up from the origin's table.

§3 PRICING

- **Distance** = rooms flown, or the wrap-aware Manhattan distance; they agree to within 3 rooms.
- **Currency**. The current code stores platinum and charges `cost * 1000` coins (`coin_stringv(cost * 1000)`).
- **The formula is Fotenak's decision**. Some illustrative linear rates:

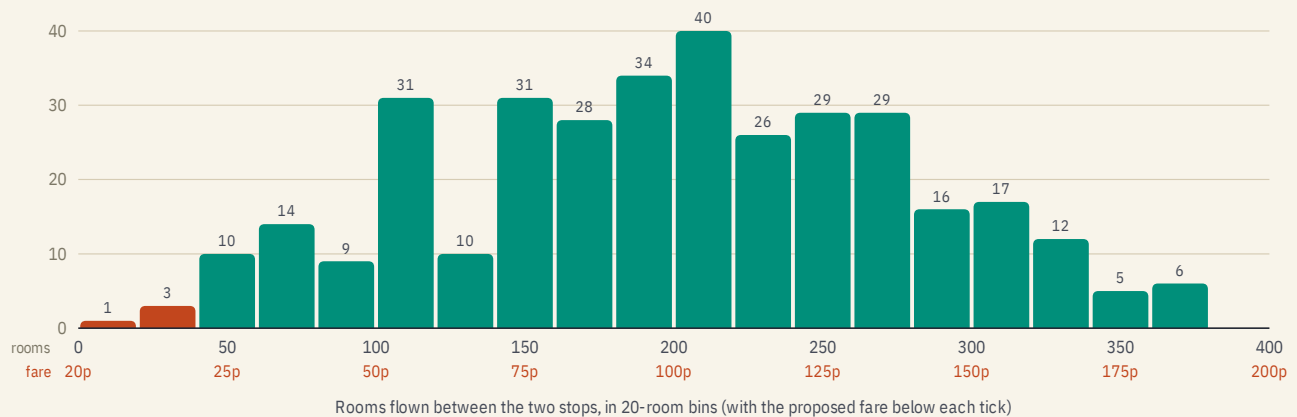
RATE	12 ROOMS (SHORTEST)	201 ROOMS (MEDIAN)	379 ROOMS (LONGEST)
0.10 p/room	1p	20p	38p
0.25 p/room	3p	50p	95p
0.50 p/room	6p	100p	190p

A minimum fare and a rounding rule are probably needed, since the shortest hop is only 12 rooms.

DECIDED · FOTENAK, ANSWER 1

“20p for a short flight, and 250p seems like a good starting point. 3rd option (0.5 plat per room).”

At 0.5p per room, hops under 40 rooms fall to the 20p minimum (4 of the 351 pairs). The longest hop, 379 rooms, costs about 190p, so a 250p ceiling would never be reached at this rate. What the 250p figure refers to still needs confirming.



How the 351 fares would spread under the proposed rate. Each bar counts the pairs of stops in a 20-room band. The red bar (under 40 rooms) is where the 20p minimum applies. Distances are from the issue's matrix, with Fort Boyard at 564319.

§4 TICKETS

Store what the ticket is for, not a table position.

The ticket should record the **destination** (catalog id or destination room vnum) and the origin mob vnum.

Validate on give.

The ticker's origin must be this mob, and its destination must be valid from this origin.

Hardening while in here:

- `flying_transport_cmd_give` trusts `ticket->value[7]` with no range or origin check (L540).
- Both movement events index `transport_routes[]` with only a `< 0` check (L641, L750).
- Tickets are `ITEM_NORENT` (extra flags `8388616 = NORENT | NOSELL`), and player snapshots leave `NORENT` items out (L328, L691–L696). So stale tickets shouldn't survive a relog or reboot.
- Even so, a ticket held across a hot table change, or edited by an immortal, can currently point anywhere.

§5 PER-CHARACTER PERSISTENCE

There are two options:

RECOMMENDED

A A new player component in the revisioned player snapshot.

This follows the small per-player sets `PLAYER_COMPONENT_LANGUAGES` and `PLAYER_COMPONENT_INTRODUCTIONS` (`player_revision_state.h` L12–L25). The learned set then travels in the same revisioned player snapshot as the EXP it triggers.

B A table like `player_recipes`.

It would hold (`pid`, `destination`) and be written immediately with `INSERT IGNORE` when learned (`sql_player.c`). It also needs a flat-file repository for `__NO_MYSQL__` builds, like `flatfile_recipe_repository`.

A NEW TABLE MUST BE REGISTERED EVERYWHERE `PLAYER_RECIPES` IS:

- ☐ an immutable migration under `migrations/immutable/` (additive, guarded, re-runnable), plus `migrations/migration_manifest.json`;
- ☐ `migrations/runtime_compatibility_manifest.json` and `src/core/runtime_compatibility_contract.h` (table list and count);
- ☐ `migrations/data_lifecycle_manifest.json` (export rule, season action, retention);
- ☐ the pwipe (`sql.c` L4990–L5000) and `sql_verify_pwipe_manifest()`;
- ☐ character deletion's `pid_deletes`;
- ☐ personal-data export and erasure;
- ☐ the flat-file mode;
- ☐ the bootstrap schema files.

Also decide what a pwipe does to learned destinations. `player_recipes` uses `season_action: reset_delete`.

DECIDED · FOTENAK, ANSWER 6

“Yes, per character, reset at pwipe.”

§6 DISCOVERY EXP

SUPERSEDED IN PART · FOTENAK, ANSWER 2

“lets scale it with the gained exp cap, and allow rested bonus to affect it. No race barriers”

This replaces the `10k × n` reading below with a reward sized from the per-gain EXP cap. The exact fraction is still to be chosen, and it needs confirming whether “no race barriers” means skipping the race EXP factor. The recorded analysis follows.

Schedule. Reading "10k for each" and "goes up with the more you find" as *reward for the n-th destination = 10,000 × n*:

DESTINATIONS FOUND	REWARD FOR THAT ONE	RUNNING TOTAL
1st	10,000	10,000
5th	50,000	150,000
10th	100,000	550,000
16th	160,000	1,360,000
20th	200,000	2,100,000
27th (all)	270,000	3,780,000

Read as a flat 10k each instead, all 27 destinations give 270,000.

Grant path. Quest rewards go through `gain_exp(ch, NULL, amount, EXP QUEST)` (`quest.c L303–L315`). That path changes the amount in several ways:

- the race factor (`exp.factor.<race>`: e.g. Human 1.3, Orc 1.15, Lich and Illithid 0.1) (`L1516–L1527`);
- Rested ×1.5 or Well-rested ×2 (`L1221–L1230`);
- the `DIFFICULTY_EXP_EARNED` dial (`L1533`);
- a cap of 1/3 of the next level per gain (`L1542–L1544`);
- no gain at all once current EXP reaches 2× the next level (`L1557–L1566`);
- nothing in arena, `ROOM_SAFE` or `ROOM_GUILD` rooms (`L1179`), though none of the stops are;
- immortals get nothing.

What the 1/3 cap does to $10k \times n$, from `exp.required.*`, before race, rested and difficulty modifiers. Level-ups are capped at `exp.maxExpLevel=50`.

LEVEL	NEXT LEVEL NEEDS	MAX PER GAIN	10K × N PASSES UNCAPPED	10K AS % OF THAT LEVEL
1–4	2,000	666	never	500%
5–9	8,000	2,666	never	125%
10–14	25,000	8,333	never	40%
15–19	100,000	33,333	$n \leq 3$	10%
20–24	400,000	133,333	$n \leq 13$	2.5%
25–29	1,600,000	533,333	all 27	0.62%
30–34	3,000,000	1,000,000	all 27	0.33%
35–39	6,000,000	2,000,000	all 27	0.17%
40–44	12,600,000	4,200,000	all 27	0.08%
45–49	20,000,000	6,666,666	all 27	0.05%
50	40,000,000	13,333,333	all 27	0.03%

A flat amount is huge for new characters (and gets capped) and negligible from level 30 on. Quest EXP rewards are capped at 1/10 of the next level per reward (`quest.c L311–L314`), so a level-relative formula may fit better. That is Fotenak's call.

Grant it exactly once per character per destination, even across a crash. Either save the learned destination first and grant EXP only after that succeeds, or commit both in one revisioned save (option A). Log it the way quest rewards are logged (`statuslog`, and `sql_log(..., QUESTLOG, ...)`).

Telemetry. `EXP QUEST` reports as source `quest`, reason `earned`. A dedicated `EXP_*` type would also need changes to `progression_source_for_type()` / `progression_reason_for_type()` and a new `telemetry_progression_source` value.

§7 UX AND TEXT

- **Learn message.** Something like "You have learned the flight destination: <ANSI name>!", followed by the EXP line.
- **List formatting.** Pad the ANSI names (`pad_ansi`). Keep the numbering identical between `list` and `buy` . Pick a sort order (see the open questions).
- **Sign.** [Sign 420](#) currently reads "Type 'list' when carrier is present to see available routes." Update it to mention learning destinations.
- **Help.** There is **no help entry** for the flight service today. Add one, e.g. `help flight paths` .

§8 RECOVERY AND COPYOVER

- Keep the `transport_snapshot` meaning: origin and destination room vnums. Then copyover `v13` and Redis `TRN1` need no format change; `transport_restore()` simply resolves the pair through the new model.
- Moved mobs are already handled: a snapshot whose origin no longer matches is landed at the new home.
- Update `docs/operations/WORLD_SINGLETON_RECOVERY.md` .

§9 TESTS

Existing singleton test.

`tests/async/test_world_singletons.py` compiles slices of `transport.c` cut at the markers `// Handles list command` and `int do_simple_move_skipping_procs` . Keep those markers or update the slicing.

Existing harness.

`tests/async/world_singletons_harness.cpp` hard-codes the current table: rooms `543662/519043` and mobs `47030/47028` . It also sets `TRANSPORT_ROUTE(flight) = 16` , which is Kimordril → Fort Khoralator. Update both for the new catalog.

New focused tests:

- ☐ catalog validation: every pad exists, all pads are in one zone, every pair is reachable;
- ☐ the price function;
- ☐ learn-on-`list` is idempotent, and the EXP is granted exactly once, including across a simulated crash and replay;
- ☐ ticket validation: wrong origin, unknown destination, out-of-range values;
- ☐ a persistence round trip in SQL and flat-file modes;
- ☐ pwipe, deletion and export coverage for the new table or component.

Gameplay journey with the test account: learn two stops, fly between them, check the price and the EXP, relog, and confirm the list persisted.

§10 SUGGESTED PHASING

1	2	3	4
CATALOG AND PRICING	LEARNED DESTINATIONS	DISCOVERY EXP	HELP, SIGN TEXT AND DOCS
The catalog, per-stop paths, distance pricing, destination-based tickets and Extra Destinations, with every destination visible and nothing persisted yet. This also fixes the placement drift.	Persistence plus learn-on- <code>list</code> .		

ON A STOPGAP

A stopgap that only realigns `transports[]` with the mob file is possible, but on its own:

- it strands Cryoss (47030) at Bone Fort Inn, which has no routes;
- the Fort Boyard route still departs from 563522, not the new 564319 pad;
- the Venan"Trut ↔ Torrhan leg still uses the retired 580080 pad.

It is better folded into phase 1.

Amended: Fotenak moved Fort Boyard back to 563522 (answer 10). The second point above, about the Fort Boyard route departing from 563522, no longer applies.

The Builder Answers

On 23 September Fotenak answered all eleven open questions. Each is shown with the question, the answer as written, and what it changes.

1

PRICE FORMULA.

CONFIRM DETAIL

Rate per room, minimum and maximum fare, rounding. Are Extra Destination costs in platinum?

FOTENAK — 20p for a short flight, and 250p seems like a good starting point. 3rd option (0.5 plat per room)

Rate: 0.5p per room with a 20p minimum. 4 pairs are under 40 rooms and would pay the minimum. The longest hop costs about 190p, so a **250p** ceiling would never be reached at this rate. **Confirm** what the 250p figure means. Extra Destination costs stay in platinum, as all prices are today.

2

EXP SCHEDULE.

CONFIRM DETAIL

- Does the n-th destination give $10k \times n$, or a flat 10k?
- Should it scale with level instead, given the 1/3-level cap and how small 10k is past level 30?
- Do the Rested bonus and race factors apply?

FOTENAK — lets scale it with the gained exp cap, and allow rested bonus to affect it. No race barriers

Replaces the $10k \times n$ schedule. Size the reward from the per-gain cap (1/3 of the next level), then apply Rested $\times 1.5$ or $\times 2$. **Confirm** whether "no race barriers" means skipping the race EXP factor. The fraction of the cap is still open.

3

WHAT DOES `Quest? = Y` MEAN **DECIDED**

for the 7 named dragons: Krrlystalarryn, Corrosyville, Azerathrax, Cryoss, Aurumax, Verdax and Glimmyris? Do they need dialogue or quest entries in `vehicles.qst` (none exist today)? Or should their destination be earned through a quest rather than by typing `list`?

FOTENAK — *Y* for the seven named dragons will be a `vehicles.qst` where they reward player for eliminating the others. (Note: make sure they are killable, leave corpses, etc)

New scope: a `vehicles.qst` where each named dragon asks the player to kill the others, so all seven must be killable and leave corpses. Flight mobs are spawned only by `initialize_transport()` at boot, not by zone resets. A killed dragon therefore leaves its stop without service until the next boot, unless a respawn rule is added.

4

RESTRICTIONS.

CONFIRM DETAIL

- The old table gated the Venan Trut network at levels 46/47 and the two one-way specials at 51. The sheet has no level, race or racewar limits on network destinations. Is every destination open to everyone, including flights into the other side's cities?
- Should immortals see every destination?

FOTENAK — lets remove all restrictions to start and let the dynamic ticket pricing be the only wall. Can revisit this later if it becomes problematic.

No gates on network destinations, including flights into the other side's cities. The fare is the only barrier. **Confirm** whether this also removes the level-51 gates the sheet sets on two Extra Destinations.

- 5** **MANSHAKA.** **DECIDED**
Hide it until Dalvik's Dockyard is renumbered and added to `areas/AREA`, or ship the landing now?
- FOTENAK** — *keep the surface vnum landing pad, but OK to rename the destination "Dalvik's Dockyard" until Manshaka is done being written.*
- Pad 634060 stays.** The destination is labeled *Dalvik's Dockyard* until Manshaka is written. Flights still land on the beach at 634060 until `dalvik.wld` is renumbered and added to `areas/AREA`.
-
- 6** **PERSISTENCE SCOPE.** **DECIDED**
Per character (assumed)? Reset at pwipe, like recipes?
- FOTENAK** — *Yes, per character, reset at pwipe.*
- Per character, wiped at pwipe.** Option A or B both work. As a table it would use `season_action: reset_delete`, like `player_recipes`.
-
- 7** **list DISPLAY.** **DECIDED**
 - Unknown destinations: hidden (assumed) or shown as locked?
 - Sort order: by distance, alphabetical, or sheet order?
- FOTENAK** — *sort by distance! Cool idea!*
- Sort by distance** from the current stop. Rooms flown and the wrap-aware Manhattan distance agree to within 3 rooms. Unknown destinations stay hidden, as the question assumed.
-
- 8** **KNOWN-DESTINATIONS COMMAND.** **DECIDED**
Should players be able to see their destinations anywhere (e.g. a `flights` command), or only at a flight mob?
- FOTENAK** — *Let them see it from anywhere with "flights" command*
- New command:** `flights`, usable anywhere, lists the destinations a character knows. It should get its own help entry.
-
- 9** **CANONICAL NAMES** **DECIDED**
for 47038 and 47041 (sheet vs. mob file), and the 47015 description.
- FOTENAK** — *fixed sheet to match, Corrosyvile description i will expand when I submit the quests part.*
- Canonical names are the mob-file names** for 47038 and 47041. The 47015 description will change with the quest work.
-
- 10** **RETIREMENTS AND PRICES.** **DECIDED**
Confirm retiring the 580080 (Torrhan) and 563522 (Fort Boyard) pads, and the new Extra Destination prices.
- FOTENAK** — *Torrhan i moved a few rooms to new pad 582084. Fort Boyard you are correct this was on the beach, so I moved it back to 563522 on the sheet*
- Only 580080 retires.** Torrhan uses 582084. Fort Boyard goes back to **563522**, 5 rooms from 564319. Chapter VI gives recomputed distances.
-
- 11** **"BEHEMOTH HERDERS".** **DECIDED**
The "old routes" tab ends with a row that names only "Behemoth Herders". It has no mob, rooms or price, and it is not in the new tab. That zone exists and is built: #943 `herders`, "The Behemoth Herders", entered from map room 594610 at (210, 236). The flight code has never referenced it. Is it a planned destination that still needs a mob and a pad?
- FOTENAK** — *Behemoth Herders I was considering as a pad, but ultimately went with nearby Varathorn Keep. Removed from sheet.*
- No new pad.** Varathorn Keep (47039) covers the area, and the row is gone from the sheet.

Questions are as recorded in the issue. Answers are quoted from Fotenak's comment, without the topic label some of them begin with ("Price:", "Exp Bonus:" and so on). The notes are added by this record. A *confirm detail* stamp means part of the answer can be read more than one way.

AMENDMENTS TO THE RECORD

Fotenak edited the spreadsheet while answering, so parts of the issue's data are now out of date. The issue's tables and the local CSV (`tmp/new-flight-paths.csv`) predate these edits. **Re-export the sheet before implementing.**

ITEM	IN THE ISSUE (22 SEPT)	NOW (23 SEPT)	ANSWER
Fort Boyard pad (47023)	564319, "A Beach of Fine Sand", (319, 160)	563522, "Foothills of the Mountains of Ciuto", (322, 158). Distances recomputed in Chapter VI.	10
Retired pads	580080 (2nd Torrhan pad) and 563522 (old Fort Boyard pad)	Only 580080. Torrhan moves a few rooms to 582084.	10
47038 destination name	Manshaka (no town built yet)	Dalvik's Dockyard until Manshaka is written. Pad 634060 kept.	5
47038 mob name on sheet	"a great white giant eagle"	Matches the mob file: "a mighty great eagle".	9
47041 mob name on sheet	"Glimmyris, the great faerie dragon"	Matches the mob file: "Glimmyris, the ancient faerie dragon".	9
"Behemoth Herders" row	Unexplained row on the old routes tab	Removed from the sheet. Varathorn Keep was chosen instead.	11
Level gates	46 and 47 on the Venan'Trut network; 51 on two one-way trips	No restrictions to start. The fare is the only barrier.	4
Discovery EXP	10k × n, or a flat 10k (proposed reading)	Scaled with the per-gain EXP cap. Rested applies. No race barriers.	2

AN ILLUSTRATION: A FIRST VISIT TO STORM PORT

To show how the answers fit together: a traveler who has found the other 26 stops reaches Storm Port for the first time and types `list`. The layout reuses today's `flying_transport_cmd_list` format. Destinations are sorted by distance (answer 7) and priced at 0.5p per room, rounded up, with a 20p minimum (answer 1). The Stronghold keeps its fixed top price.

ILLUSTRATION, NOT IMPLEMENTED · STORM PORT PAD 588326 · MOB 47017

```
< 2104h/2104H 811v/811V > list
You have learned the flight destination: Storm Port!
(discovery EXP line - amount set by the cap-scaled rule)
Flight Service
The following routes are available:
1) Shady Grove - 22 platinum 43 rooms
2) Fort Boyard - 33 platinum 66 rooms
3) Bloodstone Keep - 55 platinum 109 rooms
4) Varathorn Keep - 58 platinum 116 rooms
5) Sarmiz'Duul - 68 platinum 135 rooms
6) Tharnadia - 72 platinum 143 rooms
7) Moonshae Island - 86 platinum 171 rooms
8) The City of Torrhan - 87 platinum 173 rooms
9) The Isle of Undeath - 89 platinum 177 rooms
10) Winterhaven - 92 platinum 183 rooms
11) Clan Shatter Stone - 111 platinum 221 rooms
12) Venan'Trut - 113 platinum 225 rooms
13) Nizari - 115 platinum 229 rooms
14) Bone Fort Inn - 116 platinum 231 rooms
15) Swamp of Chief Blood-Eye - 117 platinum 234 rooms
16) Helgor Outpost - 122 platinum 243 rooms
17) Kimordril - 124 platinum 247 rooms
18) Dalvik's Dockyard - 125 platinum 249 rooms
19) Fenaline - 125 platinum 250 rooms
20) Thur'Gurax - 125 platinum 250 rooms
21) Fort Khoralator - 128 platinum 255 rooms
22) Canderthal Harbor - 143 platinum 286 rooms
23) Ugta - 143 platinum 286 rooms
24) Quietus Quay - 145 platinum 289 rooms
25) The Jade Empire - 151 platinum 302 rooms
26) Myrabolus - 166 platinum 331 rooms

Extra Destination (one-way):
27) Storm Port Stronghold - 10 platinum

Type 'buy <id>' to ride to the destination
```

The grey room counts would not appear in game; they are shown here only to explain the order. The discovery EXP line is a placeholder. Fort Boyard is measured from its amended pad, 563522 (66 rooms from Storm Port). The pricing rule and every line of text here are illustrations for discussion, not part of the recorded design.

Correspondence & References

The two comments on the issue, verbatim, followed by the issue's references.

Comment 1 of 2 · moshehbenavraham · 22 September 2026 · [issuecomment-5786076560](#)

Cross-reference: #615 is the other request from the same vehicles-area update, the Stromvok ferry. It's on [enhancement/615-stromvok-ferry](#).

The two changes don't overlap in code. Ferries come from `ferries[]` in `ferryact.c`, and flight mobs come from `transports[]` and `transport_routes[]` in `transport.c`. Vnum 47018 appears in both, but in separate vnum spaces.

Object 47018 is the Stromvok. **Mob** 47018 is a flight mob: `transports[]` spawns it at 588326 today, and this design moves it to Bloodstone Keep (606262).

Comment 2 of 2 · fotenak · 23 September 2026 · [issuecomment-5797507244](#)

1. Price: 20p for a short flight, and 250p seems like a good starting point. 3rd option (0.5 plat per room)
2. Exp Bonus: lets scale it with the gained exp cap, and allow rested bonus to affect it. No race barriers
3. Quest: Y for the seven named dragons will be a vehicles.qst where they reward player for eliminating the others. (Note: make sure they are killable, leave corpses, etc)
4. Restrictions: lets remove all restrictions to start and let the dynamic ticket pricing be the only wall. Can revisit this later if it becomes problematic.
5. Manshaka: keep the surface vnum landing pad, but OK to rename the destination "Dalvik's Dockyard" until Manshaka is done being written.
6. Persistence scope: Yes, per character, reset at pwipe.
7. List display: sort by distance! Cool idea!
8. Let them see it from anywhere with "flights" command
9. Canonical names: fixed sheet to match, Corrosyvile description i will expand when I submit the quests part.
10. Torrhan i moved a few rooms to new pad 582084. **Fort Boyard you are correct this was on the beach, so I moved it back to 563522 on the sheet**
11. Behemoth Herders I was considering as a pad, but ultimately went with nearby Varathorn Keep. Removed from sheet.

REFERENCES

Builder commits: [0fa96b480](#), reverted in [1538cfcc4](#), re-landed in [4ee13487a](#). The original flight routes date to 2009 ([6a70c4453](#), [0f076dd24](#), [937300728](#)).

Related issues: #615 (the Stromvok ferry, from the same builder update), #235 and #315 (flight dragon singleton recovery).

Main code: `src/world/transport.c`, `src/world/transport.h`, `src/world/graph.h`, `src/world/graph.c`, `src/world/limits.c`, `src/world/quest.c`.

World files: `areas/{mob,obj,wld,zon,qst}/vehicles.*`, `areas/obj/heavens.obj` (sign 420), `areas/wld/surface.wld`, `areas/wld/dalvik.wld`.

ORIGINAL LINKS

Issue: github.com/Community-Duris/Duris/issues/616 · Discussion #629 (Ideas) · Spreadsheet: [new flight paths](#). Links into the code point at master [a71fdfee7](#).



The Sheet, Verbatim

Both tabs of new flight paths as recorded in the issue on 22 September, before Fotenak's edits.

“NEW FLIGHT VEHICLES” gid 1897554044 · same as tmp/new-flight-paths.csv

```
Mob Vnum,Old Mob Name,New Mob Name,Quest?,Load Room Vnum,Destination Name,Destination Name (ANSI),Extra Destination Min Level,Extra Destination Cost,Extra Destination #,Extra Destination Name,Extra Destination Name (ANSI)
47027,"Krrlytstalarryn, the ancient red dragon","Krrlytstalarryn, the ancient red dragon",Y,619771,Swamp of Chief Blood-Eye,&+GS&+gw&+Ga&+gm
&+Gp &+gof &+GChief &+rBlood-&+RE&+ry&+Re&N,51,100,620606,A Huge Cavern on the Bloodyfist Mountains,&+yA Huge Cavern &+Lon the &+rBloodyfist
&+yMountains&N
47015,huge skeletal dragon,"Corrosyville, the ancient black dragon",Y,576577,The Isle of Undeath,&+LThe Isle of &+NUndeath,51,100,579365,A Desolate Island,&+LA Desolate &+yIsland&N
47016,mighty gryphon,a huge wyvern,,635756,Quietus Quay,&+mQuietus Quay&N,,,,,
47017,a giant green dragon,a mailed wyvern,,588326,Storm Port,&+WSto&+Lim Port&N,1,10,637082,Storm Port Stronghold,&+WSto&+Lim Port Stronghold&N
47018,huge hippogryph,an armored wyvern,,606262,Bloodstone Keep,&+RBloodstone Keep&N,,,,,
47019,flying carpet,a great white hippogriff,,514072,Myrabolus,&+MMyrabolus&N,,,,,
47020,emerald dragon,a great white griffin,,582084,The City of Torrrhan,&+WThe City of &+YTorrhan&N,,,,,
47021,powerful gryphon,a huge griffin,,543701,Ugta,&+yUgta&N,,,,,
47022,hippogryph,a flying carpet,,545407,Venan'Trut,&+YVenan'Trut&N,,,,,
47023,onyx dragon,"Azerathrax, the ancient blue dragon",Y,564319,Fort Boyard,&+rFort &+RBoyard&N,,,,,
47024,a great blue drake,a large wyvern,,622276,Sarmiz'Duul,&+ySarmiz'Duul&N,,,,,
47025,a saddled fire drake,a harnessed roc,,519043,Fort Khoralator,&+WFort &+CKhoralator&N,,,,,
47026,a GREAT wyvern,a harnessed griffin,,543662,Kimordril,&+YKimordril&N,,,,,
47028,a mighty griffon,a harnessed frost drake,,522247,Helgor Outpost,&+RHel&+rgor &+WOutpost&N,,,,,
47029,a lithe black drake,a harnessed wyvern,,631413,Clan Shatter Stone,&+WClan Shatter Stone&N,,,,,
47030,a saddled white drake,"Cryoss, the ancient white dragon",Y,514279,Bone Fort Inn,&+WBone Fort Inn&N,,,,,
47031,,a mighty roc,,533479,Winterhaven,&+WWinterhaven&N,,,,,
47032,,a mighty wyvern,,584293,Shady Grove,&+GShady Grove&N,,,,,
47033,,a mighty griffin,,570024,Tharnadia,&+WTharnadia&N,,,,,
47034,,a large hippogriff,,529229,Feneline,&+cFeneline,,,,,
47035,,a large griffin,,546144,Canderthal Harbor,&+WCanderthal Harbor&N,,,,,
47036,,,"Auzumax, the ancient copper dragon",Y,575328,Nizari,&+yNizari&N,,,,,
47037,,,"Verdax, the ancient jade dragon",Y,641295,The Jade Empire,&+gThe &+GJade &+gEmpire&N,,,,,
47038,,a great white giant eagle,,634060,Manshaka,&+LM&+ra&+Ln&+rs&+Lh&+ra&+Lk&+ra&N,,,,,
47039,,a huge skeletal dragon,,577835,Varathorn Keep,&+WV&+Na&+Wr&+Na&+Wt&+Nh&+Wo&+Nr&+Wn &+LKeep&N,,,,,
47040,,a large wyvern,,616547,Thur'Gurax,&+rThur'Gurax&N,,,,,
47041,,,"Glimmyris, the great faerie dragon",Y,557220,Moonshae Island,&+GMoonshae &+YIsland&N,,,,,
47042,,,,,,,,,
```

“OLD ROUTES” gid 0 · matches transport_routes[] row for row

```
Mob Vnum,Old Mob Name,Mob Load Room,Start Room,Destination Room,Old Destination Name,Cost,Min Level,Max Level,Extra Destination?
47027,ancient red dragon,619771,619771,620606,A Huge Cavern on the Bloodfist Mountains,50,51,0,620606
47015,huge skeletal dragon,576577,576577,579365,A Desolate Island,200,51,0,579365
47016,mighty gryphon,606262,606262,635756,Quietus Quay,20,0,0,No
47016,mighty gryphon,606262,606262,588326,Storm Port,20,0,0,No
47017,giant green dragon,635756,635756,606262,Bloodstone Keep,20,0,0,No
47018,huge hippogryph,588326,588326,606262,Bloodstone Keep,20,0,0,No
47021,powerful gryphon,543701,543701,514072,Myrabolus,10,0,0,No
47021,powerful gryphon,543701,543701,582084,The City of Torrrhan,10,0,0,No
47025,saddled fire drake,582084,582084,543701,Ugta,10,0,0,No
47022,hippogryph,514072,514072,543701,Ugta,10,0,0,No
47020,emerald dragon,580080,580080,545407,Venan'Trut,50,47,0,No
47019,flying carpet,545407,545407,564319,Fort Boyard,50,46,0,No
47019,flying carpet,545407,545407,622276,Sarmiz'Duul,50,46,0,No
47019,flying carpet,545407,545407,580080,The City of Torrrhan,50,46,0,No
47023,onyx dragon,622276,622276,545407,Venan'Trut,50,47,0,No
47024,great blue drake,563522,563522,545407,Venan'Trut,50,47,0,No
47030,saddled white drake,543662,543662,519043,Fort Khorcalator,100,0,0,No
```

47028,mighty griffon,519043,519043,543662,Kimordril,100,0,0,No
47029,lithe black drake,631413,631413,522247,Helgor Outpost,100,0,0,No
47026,GIANT wyvern,522247,522247,631413,Clan Shatter Stone,100,0,0,No
,,,,,Behemoth Herders,,,,No

DISTANCE MATRIX, AS TEXT

	27	15	16	17	18	19	20	21	22	23	24	25	26	28	29	30	31	32	33	34	35	36	37	38	39	40	41
47027 Swamp of Chief Blood-Eye	-	114	55	234	125	235	181	260	222	284	111	220	299	232	71	244	292	211	271	316	211	154	130	147	169	32	307
47015 The Isle of Undeath	114	-	169	177	159	261	107	158	108	173	213	210	197	206	173	258	210	135	169	266	109	52	244	261	61	130	205
47016 Quietus Quay	55	169	-	289	180	180	206	225	225	339	154	195	264	207	68	219	267	266	296	261	188	179	75	100	224	57	332
47017 Storm Port	234	177	289	-	109	331	173	286	225	67	135	255	247	243	221	231	183	43	143	250	286	229	302	249	116	250	171
47018 Bloodstone Keep	125	159	180	109	-	360	238	317	207	159	54	201	353	205	112	187	199	86	249	356	268	211	255	268	98	141	277
47019 Myrabolus	235	261	180	331	360	-	182	103	213	278	326	183	84	195	248	193	241	351	188	81	152	209	105	112	322	219	160
47020 The City of Torzhan	181	107	206	173	238	182	-	113	215	210	292	317	118	313	252	365	317	193	90	187	150	61	159	154	162	149	126
47021 Ugta	260	158	225	286	317	103	113	-	110	233	371	204	39	200	293	252	204	293	143	108	49	106	162	215	219	228	115
47022 Venan'Trut	222	108	225	225	207	213	215	110	-	159	261	102	149	98	191	150	102	183	245	218	65	154	272	325	109	238	217
47023 Fort Boyard	284	173	339	67	159	278	210	233	159	-	185	189	194	177	271	165	117	73	120	197	220	219	369	316	118	300	118
47024 Sarmiz'Duul	111	213	154	135	54	326	292	371	261	185	-	175	379	179	86	133	181	112	275	321	322	265	229	214	152	143	303
47025 Fort Khoralator	220	210	195	255	201	183	317	204	102	189	175	-	243	12	149	48	72	213	309	212	167	256	242	295	155	252	273
47026 Kimordril	299	197	264	247	353	84	118	39	149	194	379	243	-	239	332	257	209	267	104	69	88	145	189	176	258	267	76
47028 Helgor Outpost	232	206	207	243	205	195	313	200	98	177	179	12	239	-	161	52	60	201	297	200	163	252	254	307	151	264	261
47029 Clan Shatter Stone	71	173	68	221	112	248	252	293	191	271	86	149	332	161	-	173	221	198	342	329	256	225	143	160	156	103	378
47030 Bone Fort Inn	244	258	219	231	187	193	365	252	150	165	133	48	257	52	173	-	48	189	285	188	215	304	266	281	203	276	249
47031 Winterhaven	292	210	267	183	199	241	317	204	102	117	181	72	209	60	221	48	-	141	237	160	167	256	314	329	155	324	201
47032 Shady Grove	211	135	266	43	86	351	193	293	183	73	112	213	267	201	198	189	141	-	163	270	244	187	341	292	74	227	191
47033 Tharnadia	271	169	296	143	249	188	90	143	245	120	275	309	104	297	342	285	237	163	-	107	180	117	249	196	208	239	36
47034 Fenaline	316	266	261	250	356	81	187	108	218	197	321	212	69	200	329	188	160	270	107	-	157	214	186	169	315	300	79
47035 Canderthal Harbor	211	109	188	286	268	152	150	49	65	220	322	167	88	163	256	215	167	244	180	157	-	89	211	264	170	179	152
47036 Nizari	154	52	179	229	211	209	61	106	154	219	265	256	145	252	225	304	256	187	117	214	89	-	198	215	113	122	153
47037 The Jade Empire	130	244	75	302	255	105	159	162	272	369	229	242	189	254	143	266	314	341	249	186	211	198	-	53	299	114	265
47038 Manshaka	147	261	100	249	268	112	154	215	325	316	214	295	176	307	160	281	329	292	196	169	264	215	53	-	316	131	232
47039 Varathorn Keep	169	61	224	116	98	322	162	219	109	118	152	155	258	151	156	203	155	74	208	315	170	113	299	316	-	185	236
47040 Thur'Gurax	32	130	57	250	141	219	149	228	238	300	143	252	267	264	103	276	324	227	239	300	179	122	114	131	185	-	275
47041 Moonshae Island	307	205	332	171	277	160	126	115	217	118	303	273	76	261	378	249	201	191	36	79	152	153	265	232	236	275	-

How This Record Was Made

The sources, what was computed for this record, and how it was checked.

SOURCES

- **The issue and its comments:** section “#616” of `docs/ongoing-projects/2026-09-23-issues-moved-to-discussions.md`. That file holds the full issue body and both comments, copied from the GitHub API just before the move to Discussions. On 24 September 2026 the file was untracked in git, so it existed only in the local checkout.
- **World data:** `areas/wld/surface.wld` (160,004 rooms, zone #5000) and `areas/mob/vehicles.mob`, both as of master `e1357a30a`.
- **Code:** `src/world/transport.c`, `src/world/graph.h` (`VALID_FLYING_EDGE`), `src/core/common.c` (`rev_dir`) and `src/core/utility.c` (ANSI codes, `coin_stringv`).

RECOMPUTATION

This record re-derived the flight distances instead of copying them. The surface rooms and their exits were parsed, and a flight edge was allowed only where `VALID_FLYING_EDGE` allows one. That means an exit whose target is in the same zone, is not `EX_BLOCKED`, and has a matching exit back through `rev_dir`. Breadth-first search then ran from every pad.

- ✓ **Distance matrix:** 702 of 702 directed cells match the issue.
- ✓ **Current routes:** 20 of 20 “rooms flown” values match.
- ✓ **Extra Destinations:** 3 of 3 match (37, 19 and 166 rooms).
- i **New:** Fort Boyard measured from 563522, the pad in answer 10. It is 5 rooms from 564319, and its distances to the other 26 stops are in the last row of the Chapter VI matrix.

THE MAPS

The maps are rendered from the sector type of each surface room. Ocean depth comes from the distance to land, relief from a height value per sector (fields low, hills higher, mountains highest), and some colors from room names, such as snow, jungle, sand, bog and lava. The routes on the maps are stylized curves between stops, not the exact paths flown. Every curve takes the short way around the wrapping world.

Destination and mob names are drawn from their ANSI codes: `&+x` for normal colors, `&+X` for bright, and `&N` to reset. `&+L` is dark grey and `&+y` is brown. Exact colors depend on each player's terminal.

ADDITIONS NOT IN THE ISSUE

- Every map, chart and color-rendered name.
- The recomputed Fort Boyard row, and the room title for 563522.
- The status of each builder nit, the “what it changes” notes in Chapter VIII, and the amendments table.
- The Storm Port `list` illustration, which is not a spec.